

# Assignment 7

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## Exercise 1

- a) A case primarily consists of a specific problem description and its corresponding solution. In many advanced systems, a case also includes the evaluation outcome and an explanation or justification of exactly how that solution was reached.
- b) The main processes are Retrieve, Reuse, Revise and Retain. In Retrieve, the system looks for the most similar cases. During Reuse, the retrieved case is adjusted to suggest a solution. In Revise, the proposed solution is evaluated. Finally, in Retain, the new experience is saved for future use.

## Exercise 2

- a) An application could be a meal planning assistant. The task of the system is to recommend a cooking recipe based on the ingredients a user currently has in their kitchen.
- b) The case representation includes the main ingredient, secondary ingredients, dietary restrictions, cooking time, and the specific recipe provided as the solution.
- c) A similarity function for the main ingredient attribute yields 1 for an exact match, 0.5 for an ingredient in the same food group, and 0 for completely different foods.

## Exercise 3

|                 | Vocabulary                                                      | Case Base                                                         | Similarity                                                              | Adaptation                                                         |
|-----------------|-----------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Retrieve</b> | Specifies the language used to represent the problem.           | Acts as the repository searched for similar past cases.           | Guides the retrieval of experiences most similar to the target problem. | Not primarily used in this step.                                   |
| <b>Reuse</b>    | Specifies the language used to represent the proposed solution. | Provides the retrieved cases to be utilized.                      | Not primarily used in this step.                                        | Modifies the retrieved solution to address the new target problem. |
| <b>Revise</b>   | Represents the evaluation and repaired knowledge.               | Provides the original proposed solution for domain expert review. | Not primarily used in this step.                                        | Explains and repairs failures in the proposed solution.            |
| <b>Retain</b>   | Can be revised as the system learns and evolves.                | Retains the new case if it is deemed useful.                      | Can be updated as knowledge shifts during system evolution.             | Can be updated as knowledge shifts during system evolution.        |